

Identification of cheese mite species inoculated on Mimolette and Milbenkase cheese through cryogenic scanning electron microscopy.

Samples of Mimolette (France) and Milbenkase (Germany) cheeses traditionally ripened by mites were analyzed to determine the mite species present on each sample. Scientific literature was reviewed to understand which mite species most commonly infest cheese. Morphological features possessed by mites were then studied to understand what unique characteristics are required to ensure accurate identification. After identification and compilation of a detailed key of stored food mites (subclass Acari, order Astigmata) and their delineating features, the mites were viewed through a cryogenic scanning electron microscope. It was determined that Mimolette cheese is inoculated with *Acarus siro* L. The features studied to identify this mite species included idiosomal length and shape, setae length and arrangement, leg size, placement of anus and genitals, and solenidia shape. The Milbenkase cheese is inoculated with *Tyrolichus casei* Oudemans, which was evident after viewing the same features used to identify *A. siro* and the supracoxal seta shape. With this knowledge, further research can be conducted on the 2 cheese varieties to understand what chemical, physical, and microbial changes occur within the cheeses because of mites. It is important to identify the mite species present on each cheese variety to improve our understanding of their role in creating the distinctive characteristics that set these cheeses apart from others. Copyright (c) 2010 American Dairy Science Association. Published by Elsevier Inc. All rights reserved.